Differentiation: The Product Rule

Questions 1–23

Question 1. Find $\frac{dy}{dx}$ for each of the following:

a) $y = x(x-1)^2$
b) $y = x^2(2x+7)^2$
c) $y = 4x(7-x)^4$
d) $y = (x+2)^2(3x+1)^3$
e) $y = (3-x)^2(2x+9)^3$
f) $y = \sqrt{2x+5}(4x+1)^2$
g) $y = (\frac{1}{x} - 2x)(\frac{1}{x^3} + \frac{2}{x^5})$

Question 2. Three students attempt to find $\frac{d}{dx}((x^2-1)(x^3+5))$. Identify the correct version and spot the error(s) in the other two:

Student A:

$$(x^2-1)(x^3+5) = x^5 - x^3 + 5x^2 + 5$$

$$\frac{dy}{dx} = 5x^6 - 3x^2 + 10x$$

Student B:

$$\frac{dy}{dx} = (2x)(x^3+5) + (x^2-1)(3x^2) = 5x^4 - 3x^2 + 10x$$

Student C:

$$\frac{dy}{dx} = 2x \times 3x^2 = 6x^3$$

Question 3. Differentiate the following:

a) $y = x \ln x$
b) $y = \sqrt{x} \tan x$
c) $y = \sin x \cos x$
d) $y = \cos x \ln x$
e) $y = 2^x \sin x$
f) $y = (x^2 + 1) \ln x$

Question 4. Differentiate and fully factorise the following:

a) $y = xe^x$
b) $y = x^2 4^x$
c) $y = e^x \tan x$
d) $y = 2^x \sin x$
e) $y = (x^2 - 12)e^{-2x}$
f) $y = e^{4x} \sin 2x$

Question 5. Find the equation of the tangent line to the curve at the given point, giving your answers in the form $y = mx + c$:

a) $y = x^3(x+2)^2$ at $x = 1$
b) $y = x^5\sqrt{3x+10}$ at $x = 2$
c) $y = (x^2 - x + 4)e^x$ at $x = 0$

Question 6. Fully factorise each of the following:

a) $4x^2 + 2x^2(x+1)$
b) $3(2x-1)^5 - (x+1)(2x-1)^4$
c) $4(3x+1)^3 + (x+2)(3x+1)^2$
d) $\frac{1}{3}(6x+5)^3 + (x-1)(6x+5)^2$
e) $\frac{1}{2}(1-x)^{-5} + 4(1-x)^{-6}$
f) $4(4x+1)^3(3x-1) + 9(3x-1)^2(4x+1)^2$

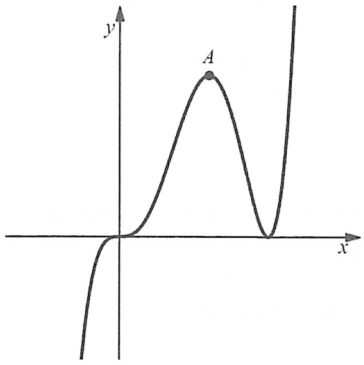
Question 7. Solve each of the following:

a) $2(x+1)^3 + (x-1)(x+1)^2 = 0$

b) $\frac{1}{3}(2x+1)^{-2} + (2x+1)^{-3} = 0$

c) $5(7x-1)^3(x-1) - 3(x-1)^2(7x-1)^2 = 0$

Question 8. The curve shown below is $y = x^3(3x-5)^2$.



A local maximum value is at the point A.

a) Find the coordinates of A.

b) State the coordinates of the other stationary points.

c) State the values of y for which the $y = x^3(3x-5)^2$ has three unique solutions.

Question 9. For each of the following, differentiate and fully factorise:

a) $y = x(x+3)^7$
b) $y = (x-3)(2x+1)^4$
c) $y = 5x(2x+1)^4$
d) $y = (1+x^2)e^{2x}$

Question 10. A curve is given by $y = 4000xe^{-2x}$.

a) Find the stationary point of the curve.
b) Determine its type.

Question 11. The number of employees y, in a company in the first 12 months may be modelled by $y = 150 - 3t\sin(\frac{\pi}{6}t)$ where t is the number of months after 1st January.

a) Find the number of employees predicted by the model on 1st January.
b) Find an expression for $\frac{dy}{dt}$
c) Find the rate of change in the number of employees predicted by the model on 1st April.
d) Interpret your result in part c) giving your answer in context.

Question 12. Find the equation of the tangent line to the curve $y = x^4e^{2x}$ when $x = 1$. Give your answer in the form $y + pe^2x + qe^2 = 0$

Question 13. A curve is given by $y = x \ln x$, $x > 0$. Find the stationary point and determine its type.

Question 14. A curve is given by $y = e^{-x} \sin x$, $x \geq 0$.

a) Show that the curve has a stationary point at $(\frac{\pi}{4}, \frac{\sqrt{2}}{2} e^{-\pi/4})$

b) Determine its type.

Question 15. Prerequisite knowledge for Q15: Geometric Sequences A curve is given by $y = 10e^x \cos x$, $x \geq 0$.

a) Show that the x coordinates of the stationary points of the curve satisfy the equation $\tan x = 1$

The y-values of the stationary points satisfy a geometric sequence,

b) find the value of the common ratio r.

Question 16. Find $\mathrm{f}'(x)$ for each of the following:

a) $\mathrm{f}(x) = e^{10x^2} \sin(2x^3)$

b) $\mathrm{f}(x) = 5^{x^2} \cos(1 - x^2)$

c) $\mathrm{f}(x) = (x^2 + 1)^3 \ln(x^2)$

Question 17. Prerequisite knowledge for Q17: Points of Inflection Show that $y = 2^{3x^2}$ is convex (concave upwards) for all values of x

Question 18. Show that the curve $y = (5 - 2x)\ln(x + 1)$ has a stationary point at $x \approx 1.04$

Question 19. For each of the following curves find $\frac{dy}{dx}$, find the coordinates of any stationary points.

a) $y = x^2e^x$

b) $y = (3x-1)^4(3x^2+1)^5$

c) $y = \sqrt{x}(x-1)^2$, $x > 0$

Question 20. For the curve given by $x = (y+2)\ln(3y-5)$, $y > \frac{5}{3}$

a) Find $\frac{dx}{dy}$ in terms of y

b) Hence find the value of $\frac{dy}{dx}$ when $x = 0$

Question 21. A curve has the equation $y = x^2 \tan^3 x$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$. Show that the gradient of the curve at $x = \frac{\pi}{4}$ is $\frac{\pi}{8}(3\pi + 4)$

Question 22. The height, h metres, from the ground of a bungee jumper is modelled by the equation

$$h = 10 + 25e^{-0.05t}(1 + \cos(\tfrac{\pi}{3}t))$$

where t is the time in seconds after jumping.

a) State the height above the ground of the start position.
b) Find $\frac{dh}{dt}$ and the value of $\frac{dh}{dt}$ when $t = 3$
c) Explain what is happening to the bungee jumper at precisely 3 seconds.

Question 23. Given that $f(x) = x\sqrt{4x+3}$, $x > -\frac{3}{4}$

a) Find $f'(x)$

b) Show that $f''(x) = \frac{12(x+1)}{(3+4x)^{3/2}}$

c) Find the x-value of the turning point and determine its nature.